DE Shaw Prop Trading Intern (Summer 2026) Case Study

Lokkit Sanjay Babu Narayanan

April 2025

In a world where waste isn't just a byproduct but a direct cost for the end user, consumer behavior would shift dramatically. One of the most fascinating knock-on effects would be in how companies design their product packaging. Every gram of material now becomes part of a personal cost optimization problem — not just for companies, but for individuals.

Firms whose core business revolves around e-commerce — such as Amazon, eBay, and others — would face intense pressure from consumers aiming to minimize their personal waste taxes. Packaging would shift from an afterthought to a critical differentiator in purchase decisions. For many users, especially in price-sensitive markets, the perceived waste burden of a product could tip the scale between brands.

This shift is especially consequential given the scale of the industry: the global e-commerce market is currently valued at \$18 trillion, with projections showing potential growth to over \$75 trillion within the next decade. Notably, the Asia-Pacific region leads in e-commerce volume — a market that is not only highly price-sensitive, but also fragmented across multiple regional players, in contrast to the relatively consolidated markets in North America and Europe. In this environment, packaging becomes not just logistics, rather a variable in consumer-facing pricing strategy.

1 Assumptions

- Consumers are charged a waste tax of \$0.50 per kilogram, multiplied by a material specific multiplier (M), based on the material's potential environmental harm.
- Heavier or more protective packaging increases product survivability, reducing the potential of damage to the product, but however increases the tax burden.
- Survivability $S(W)$ of a product increases with packaging weight, following a sigmoid function to mimic the diminishing returns of adding more material.
- If a product is damaged, the seller loses the product's value (V). Expected loss can be modeled by $(1 - (S(W)) \cdot V)$.
- Consumers are less likely to purchase from a seller when the Total Effective Cost (TEC) is higher. Conversion Rate of Sales (CR), drops linearly as TEC grows.
- The Seller's goal is to maximize expected revenue per unit.
- Revenue is modeled by $Revenue = CR \cdot (P - ExpectedLoss)$
- Conversion Rate is modeled by $CR = max(0, 1 - \lambda \cdot (TEC - P))$, where λ is a customer sensitivity coefficient, as consumer sensitivity to price varies from region to region.
- Survivability is modeled by $S(W) = min(S_{max}, S_0 + k \cdot (log(W + \epsilon))$, where k is just a coefficient that models how quickly survivability grows with regards to added weight, and ϵ is just a small constant to prevent $log(0)$.

2 Simulation Methodology

To quantify the optimal packaging choice for different product types, we simulate the expected revenue across packaging configurations for four common consumer products: a smartphone, a pair of headphones, a t-shirt, and a pack of paper towels. For each product, we analyze the tradeoff between packaging weight (ranging from 0.05–0.25 kg), environmental harm (M), survivability $S(W)$, and the effect on total cost and customer conversion.

Five packaging materials were considered:

- Styrofoam ($M = 2.0$)
- Plastic Film ($M = 1.5$)
- Recycled PET ($M = 1.0$)
- Cardboard ($M = 0.5$)
- Compostable Fiber ($M = 0.2$)

We compute for each configuration:

- Expected loss from damage: $(1 - S(W)) \cdot V$
- Waste tax burden: $W \cdot M \cdot 0.5$
- Total Effective Cost: $TEC = P + \text{Tax}$
- Conversion Rate: $CR = 1 - \lambda \cdot (TEC - P)$
- Expected Revenue: $CR \cdot (P - \text{Expected Loss})$

3 Results

The table below summarizes the optimal configuration (material and weight) for each product based on maximum expected revenue:

Product	**Material**	**Weight (kg)**	**Survivability**	**Waste Tax (\$)**	**TEC (\$)**	**Revenue (\$)**
Smartphone	Styrofoam	0.05	0.995	0.05	650.05	646.68
Headphones	Styrofoam	0.05	0.995	0.05	100.05	99.45
T-Shirt	Styrofoam	0.05	0.995	0.05	30.05	29.83
Paper Towels	Styrofoam	0.05	0.995	0.05	4.05	3.97

Table 1: Optimal Packaging Strategy per Product

4 Insights and Conclusion

The simulation reveals a powerful insight: under a waste-based tax regime, even the most environmentally harmful materials — such as Styrofoam (with a destructiveness multiplier of $M = 2.0$) — can yield the highest expected revenue when used strategically at low weights. At 0.05 kg, the resulting waste tax (\$0.05) contributes negligibly to the total effective cost (TEC), while survivability exceeds 99.5%, reducing expected product loss to near-zero. This leads to extremely high conversion rates across all product types, with expected revenue improvements of up to 25% over suboptimal packaging strategies.

Macro-Level Insight

Regulatory measures like waste taxation introduce a new form of arbitrage, one based on industry frictions between policy and adaptation. The firms that succeed are not those that simply reduce waste, but those that optimize the cost-risk-behavior equilibrium faster than competitors.

From an investment lens:

- Firms with the capability to simulate product-level tradeoffs (like TEC vs. survivability) will sustain or grow margins under regulation.
- Companies that can dynamically adjust packaging strategies by SKU, geography, and customer profile will dominate in fragmented, tax-sensitive markets.
- Material science companies that offer high-survivability, low-weight, low-multiplier packaging solutions become high-leverage acquisition targets.

Investment Strategy

Quantifying waste-tax dynamics enables a new class of trading signals:

- Long positions in firms that use high-survivability, low-weight packaging (e.g., Amazon optimizing per-SKU packaging via ML).
- Short positions in firms with high-volume, low-margin products that rely on heavier or non-optimized packaging (e.g., certain companies that focus on Fast-Moving Consumer Goods and Single-Use Packaged Goods).
- Event-driven strategies around proposed or pending regulation (e.g., front-running adoption of waste-tax policy in APAC vs. EU markets).

For instance, in our simulation, shifting from a baseline PET configuration to optimized Styrofoam yields:

- \$2–\$4 per-unit revenue gains for mid-range products like headphones or apparel.
- Up to 30–50 basis points improvement in gross margin — significant in thin-margin retail verticals.

An investment firm armed with such simulation tools could backtest packaging spend against earnings surprises, forecast competitive repositioning from cost models, and even build pricing-based consumer sentiment metrics that integrate behavioral elasticity (λ) per segment.

Conclusion

This case study highlights how a simple policy — taxing waste — can trigger complex behavioral and financial shifts across industries. By quantitatively modeling the tradeoff between packaging weight, survivability, taxation, and consumer behavior, we can identify strategies that protect margins and drive competitive advantage under regulatory pressure.

The ability to uncover these insights lies at the intersection of technology, public policy, and business strategy. Individuals and firms that understand how to navigate and simulate these domains together, using data to anticipate outcomes rather than just react to them, will be best positioned to lead in a future shaped by environmental and economic complexity.